Problem-First AI

Lecture 1:

Welcome and AI System Design



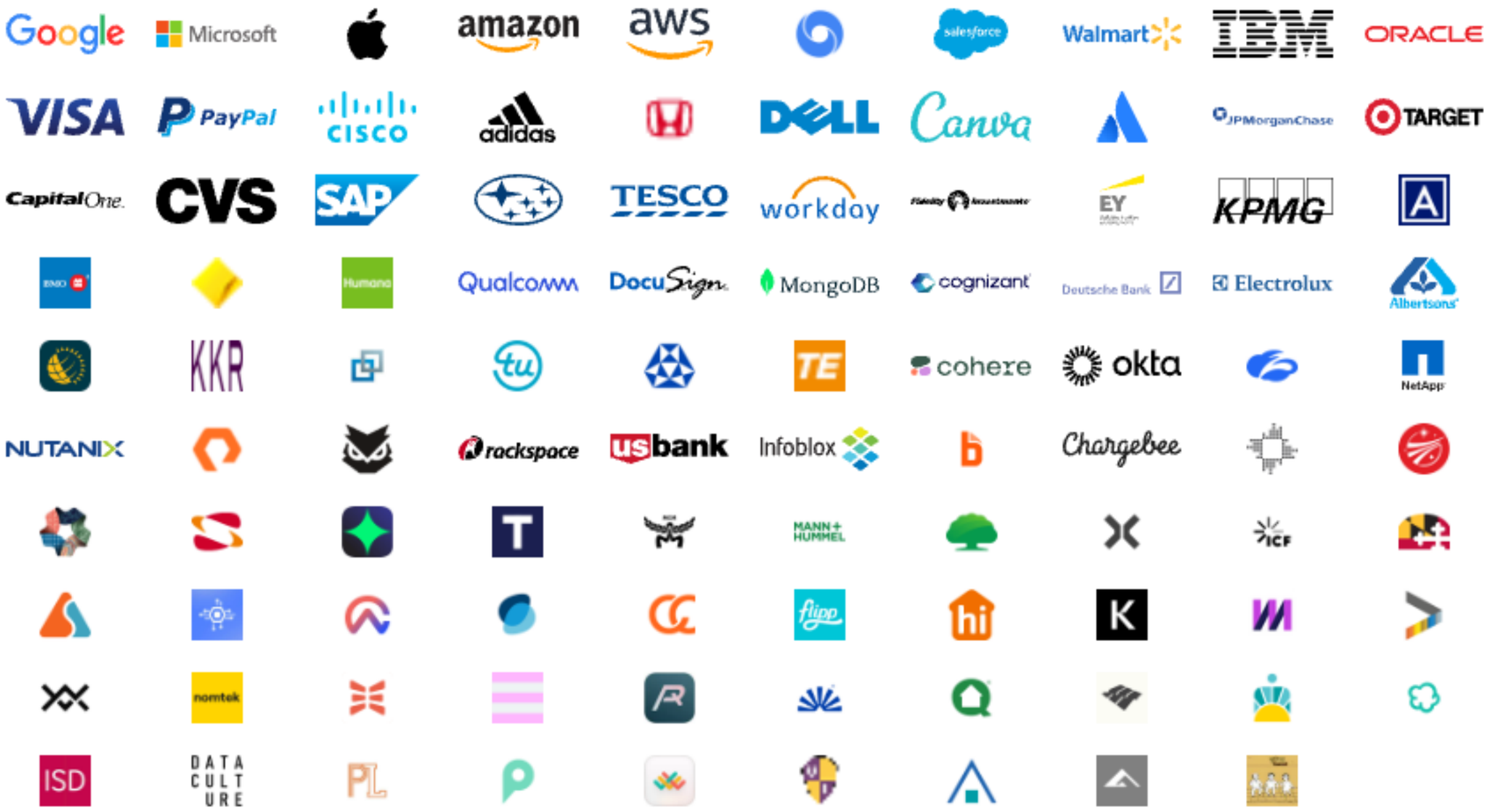
Aishwarya Reganti



Kiriti Badam

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Meet your cohort



Who's in the room?

- Product & design
- Engineering & architecture
- AI, data & research
- Business & transformation
- Founders & executives
- Consulting & customer solutions

From hands-on practitioners to senior leaders and founders.

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You’re in Good Hands

4.8/5

750+ ratings

3,200+

students

Top 5

on Maven

- Featured by Lenny Rachitsky and Maven’s CEO.
- Experience teaching private enterprise cohorts.

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Student Success

150+

students secured new AI roles

13

capstone teams building projects and raising funding

52%

of the cohort joined through referrals

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Your instructors



Aishwarya Reganti

CEO & co-founder,
LevelUp Labs, ex-AWS

- Previously at AWS GenAIIC: 35+ AI deployments, including Zillow, NFL and Strava.
- NLP researcher: 40+ papers and early work on Amazon Alexa.



Kiriti Badam

CTO & co-founder,
LevelUp Labs, ex-OpenAI

- Built AI coding agents (Codex) at OpenAI.
- Led engineering teams at Google and Samsung.
- Founding engineer at Kumo.ai.

LevelUp Labs is a human-first AI transformation lab for mid-market and enterprise companies.

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What the hell is problem-first?

Tool-first



How can we use this technology?

Problem-first



What would solve this problem?

Your problem is the hero, not AI.

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Jevons' paradox in AI

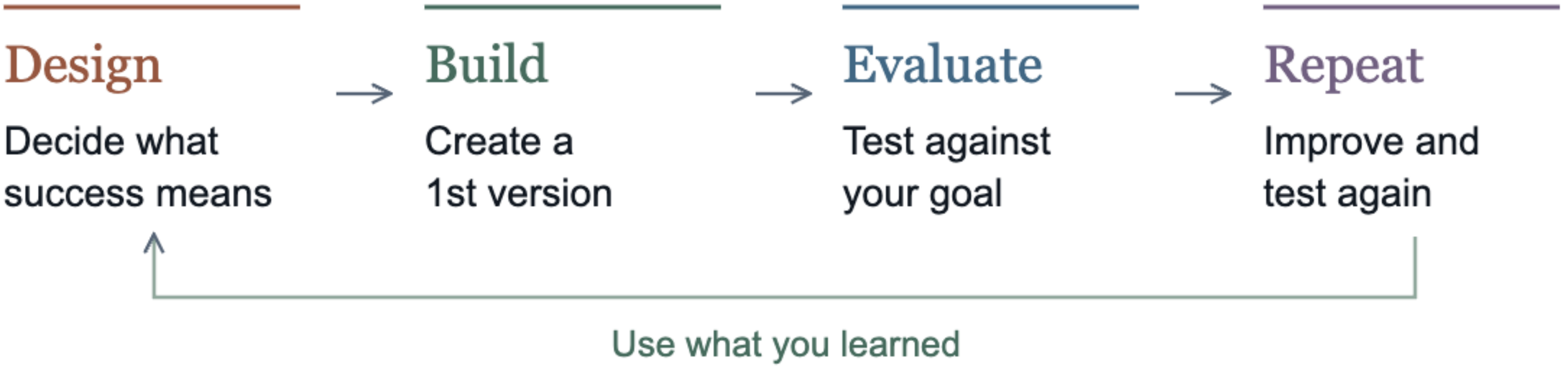


Lower cost to build → More people building

When everyone can build, choosing an impactful problem and the right tools becomes the advantage.

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The Problem-First Approach



$ Cost	Performance	Effort	Latency

"If we choose this approach, what's the trade-off?"

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Course Syllabus

Week 1	Designing and evaluating AI applications
Week 2	Context engineering: search, retrieval and memory
Week 3	Deep agents and harness engineering
Week 4	Fine-tuning, multimodal AI and AI in production
Weeks 5–6	Capstone project and demos

The goal is to unlearn old assumptions and build a mental model for designing AI systems, **not check off a list of topics.**

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How the course works



[Core] Live lectures & Slack

Learn in lectures; discuss in office hours and on Slack.



[Build] & [Grow]

Practise through [Build]. Explore [Grow] at your pace.



Capstone project

Apply the concepts to a real business problem.



Community & guests

Guest perspectives and optional Chai & AI sessions.

[Core] is taught in live lectures, with recordings to revisit. Office hours and Slack support your learning.

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Live lectures and office hours



[Core] Live lectures

2 sessions each week

Concepts, examples, and discussion

Recordings to revisit the material



Office hours

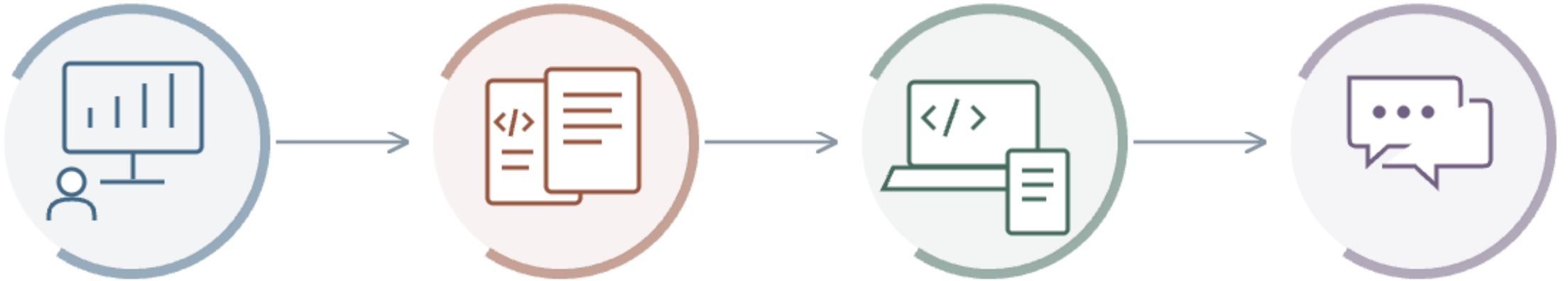
2 sessions each week

Tuesday and Thursday

Questions, implementation, and ideas

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How each week works



Live lectures

Learn through examples and discussion

Saturday

[Build] & [Grow] released after the lecture

Practise & review

Try the material and note questions

Tuesday / Thursday

Office hours
Bring your questions

Next week

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Slack, Maven, and education portal

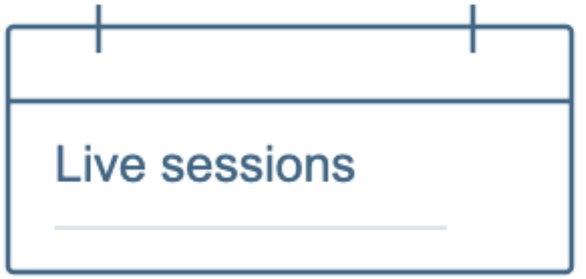
Slack



All the channels on Slack.

Next slide →

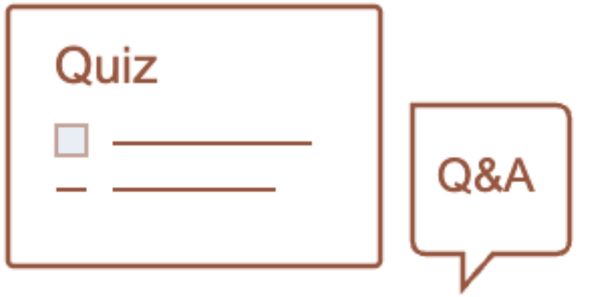
Maven



Calendar, recordings, and course materials.

Open Maven ↗

Education portal



Quizzes and live office-hour Q&A.

Open education portal ↗

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Slack support

#pfai_announcements_october_2026 ↗

Don’t post here. This is for announcements only.

#pfai_lecture_questions_october_2026 ↗

Questions about concepts from the lectures.

#pfai_homework_questions_october_2026 ↗

Help with [Build] components and implementation.

#pfai_logistics_questions_october_2026 ↗

Session invites, recordings and course access.

The teaching team aims to respond within 24 hours.

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How to Earn Points

Participate	Points
Build in public post	10
Quiz questions	1 per question
Thoughtful question in office hours	2
Most Valuable Takeaway	4

Use the education portal to follow your progress on the leaderboard.

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Build in public

Share what you're learning or unlearning through the course on LinkedIn, X, or whichever social platform you use.

If anything challenges your assumptions, share it.

The world needs less noise in AI and a more thoughtful approach.

Tag us so that we can help amplify it.



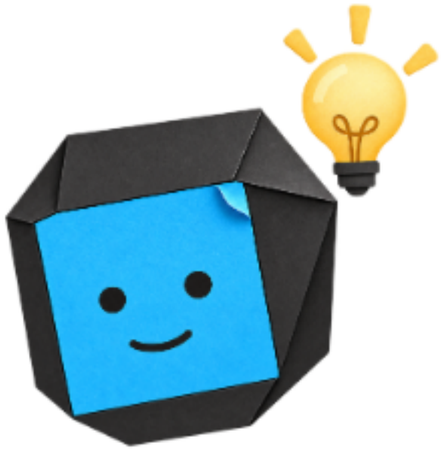
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What is the most memorable thing you learnt or unlearnt today?

Share your answer in the chat.

Vote for the takeaway that helped you most.

The most-voted reply or the instructor's favorite earns **4 points.**



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What do you get for being on top of the leaderboard?



$100 Amazon gift card

Every week, the person at the top of the leaderboard wins.

Learn, contribute, and share what you build.

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Choose your learning path

Live lectures

Learn directly through teaching, examples, and discussion.

Recordings + office hours

Study at your own pace, then bring your questions.

Both

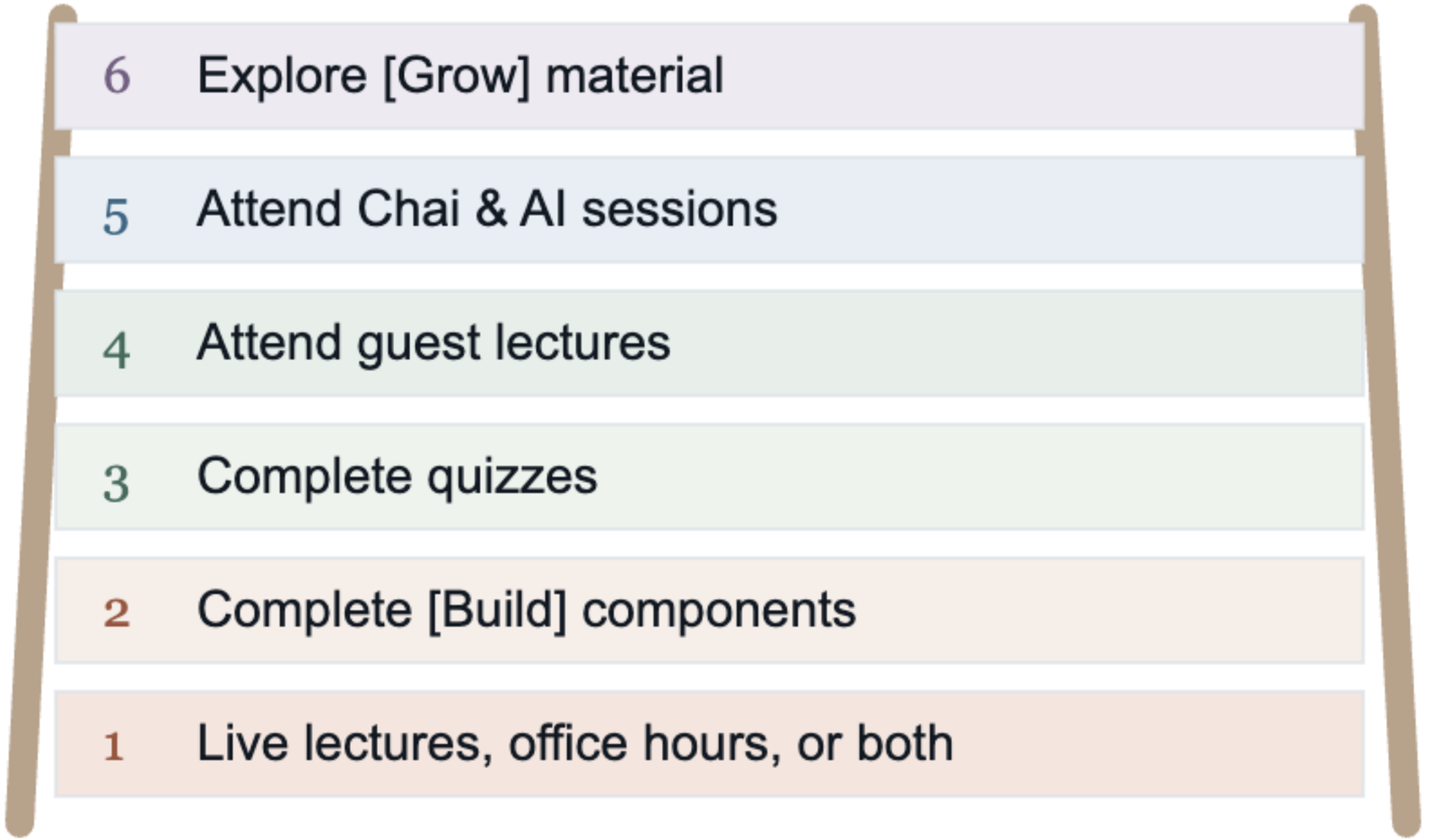
Join the teaching and return for discussion and support.

Choose what fits your schedule each week.

Recordings are up within 2–3 hours. You have lifetime access.

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Your weekly priorities



Total weekly time

8–15+ hours

Optional sessions and deeper reading

5–8 hours

Lectures, [Build], and quizzes

Review lecture material at your own pace. You have lifetime access.

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[Build] Components: LangGraph and LangFlow

LangGraph

Python framework

LangFlow

Visual, low-code framework

We chose them for building, tracing and evaluating AI applications.

Budget about $20 for OpenAI API usage for the [Build] components.

Learn the parts used in the assignments.

We'll introduce other tools when an assignment needs them.

Choose your framework ↗ Why these frameworks? ↗

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Why learn these tools if a coding agent can build it?

Claude Code or Codex can build a customer support agent.

What the application needs	What we reuse
Orchestration: Connect models and tools, pass data between them, and control what runs next.	Framework components, model/tool integrations and workflow routing.
State: Save messages, draft replies and progress so the agent can continue later.	LangGraph's saved state and pause/resume.
Observability: Inspect prompts, responses, tool calls and errors across a run.	Run inspection and integrations with tracing tools such as Arize.

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Why use AI frameworks?

Claude Code can write this infrastructure for you.
You still have to understand, debug, and maintain it.

Agent applications are changing quickly, and design practices are still developing.
Databases and web development have decades of established practice.

Frameworks reduce how much of it you build yourself.

We'd rather you spend time understanding the abstractions than writing boilerplate code.

- Choose when to use workflows, single agents or multi-agent systems.
- Write evals for real tasks and failure cases.
- Debug RAG failures in retrieval, context or generation.

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Course etiquette

Learning together

- Come in open and curious. Listen with grace.
- Leave your ego at the door. We do the same.
- Consume and contribute. Everyone brings a different piece of the puzzle.
- Keep discussions relevant. No self-promotion.
- Week 1 can be a little rough. Stick with us.

Asking questions

- Slack for quick questions; office hours for longer discussions.
- Try the basics with AI first. Come prepared.
- Focus on that week's material; submit questions ahead, with context.
- No DMs, please, pretty please! Bring your business question to the group.

Extend a little grace to the community.

Please keep your mic muted and avoid interrupting live lectures so everyone can follow; bring your questions to office hours.

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Bring good energy

Welcome to our community.
We're thrilled to have you here.



Ravi
Teaching partner

Sahana
Teaching partner

AISH
Instructor

Kiriti
Instructor

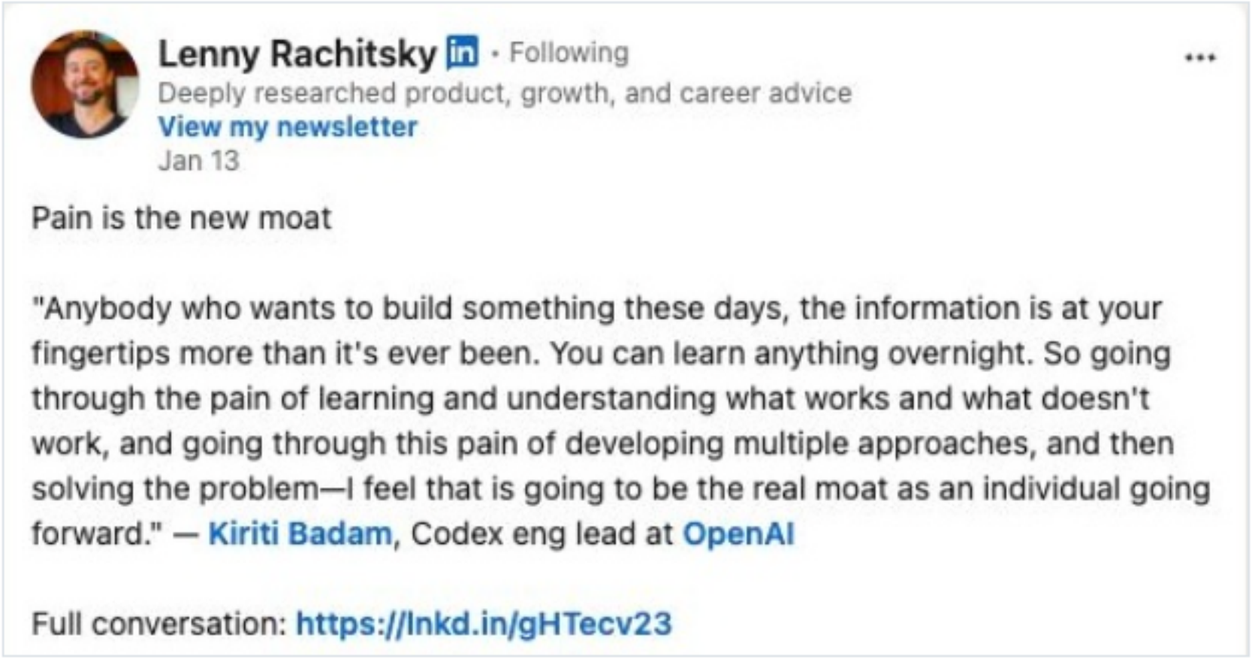
Koa
Course operations

Cielo
Course operations

Bring curiosity, contribute generously, and support each other.

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Pain Is the New Moat



Lenny Rachitsky · Following
Deeply researched product, growth, and career advice
View my newsletter
Jan 13

Pain is the new moat

"Anybody who wants to build something these days, the information is at your fingertips more than it's ever been. You can learn anything overnight. So going through the pain of learning and understanding what works and what doesn't work, and going through this pain of developing multiple approaches, and then solving the problem—I feel that is going to be the real moat as an individual going forward." — **Kiriti Badam**, Codex eng lead at **OpenAI**

Full conversation: **https://lnkd.in/gHTecv23**

This is not an easy course, but it can be the most transformational course you take.

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Lecture 1

AI System Design

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Recap: From choosing a problem to designing AI

Start with a real problem.

Choose a small, useful slice and expand what works.

Understand the pieces.

A model uses context, instructions, and tools to do the work.

Judge the whole application.

Model benchmarks are a starting point. Product evals test your use case.

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What we'll cover today

- How do we design and iterate AI solutions?
- How do we choose models and context?
- What trade-offs shape the application?
- How do we improve instructions and the 1st version?

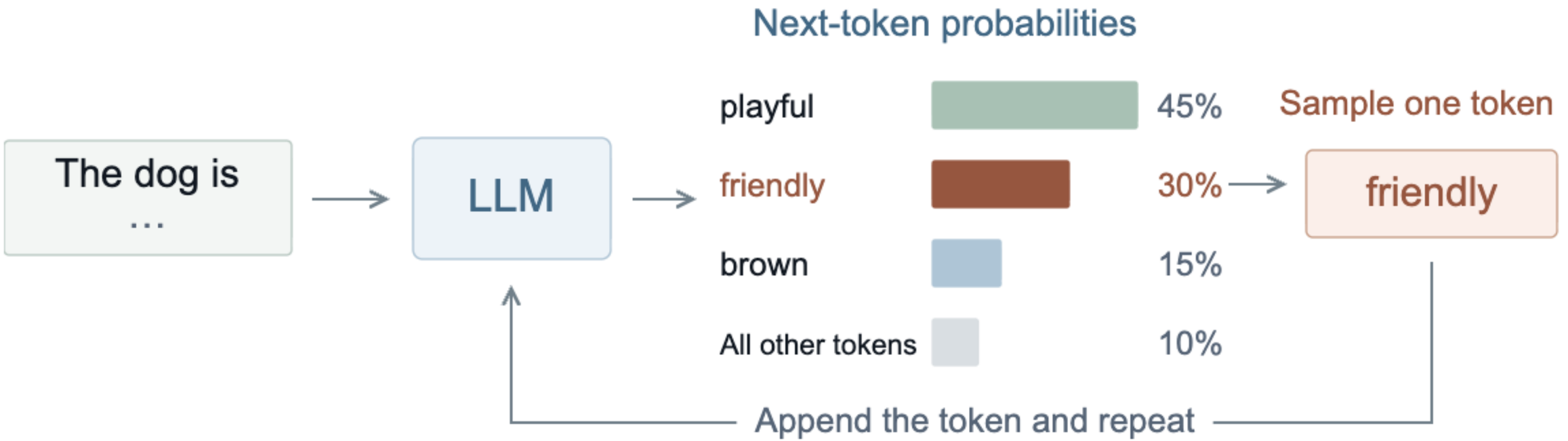
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Why does AI require a different system design?

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Problem 1: Non-determinism

An LLM assigns next-token probabilities across a large vocabulary.

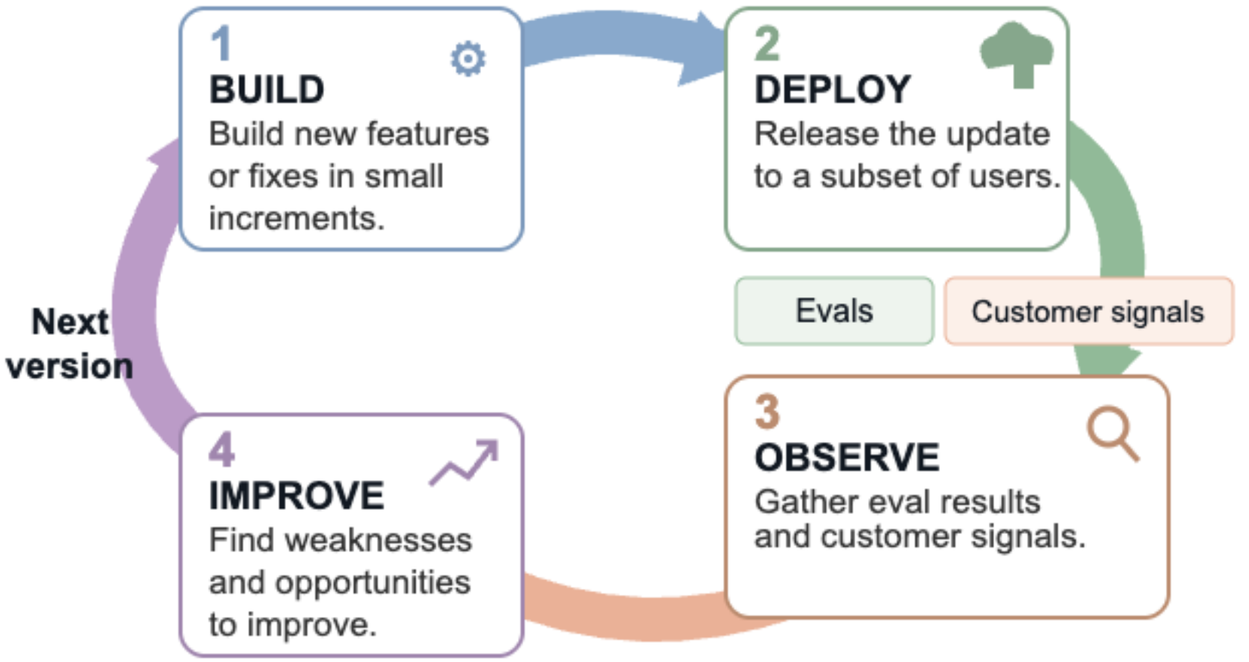


Sampling can make generation non-deterministic: the same prompt can produce different responses.

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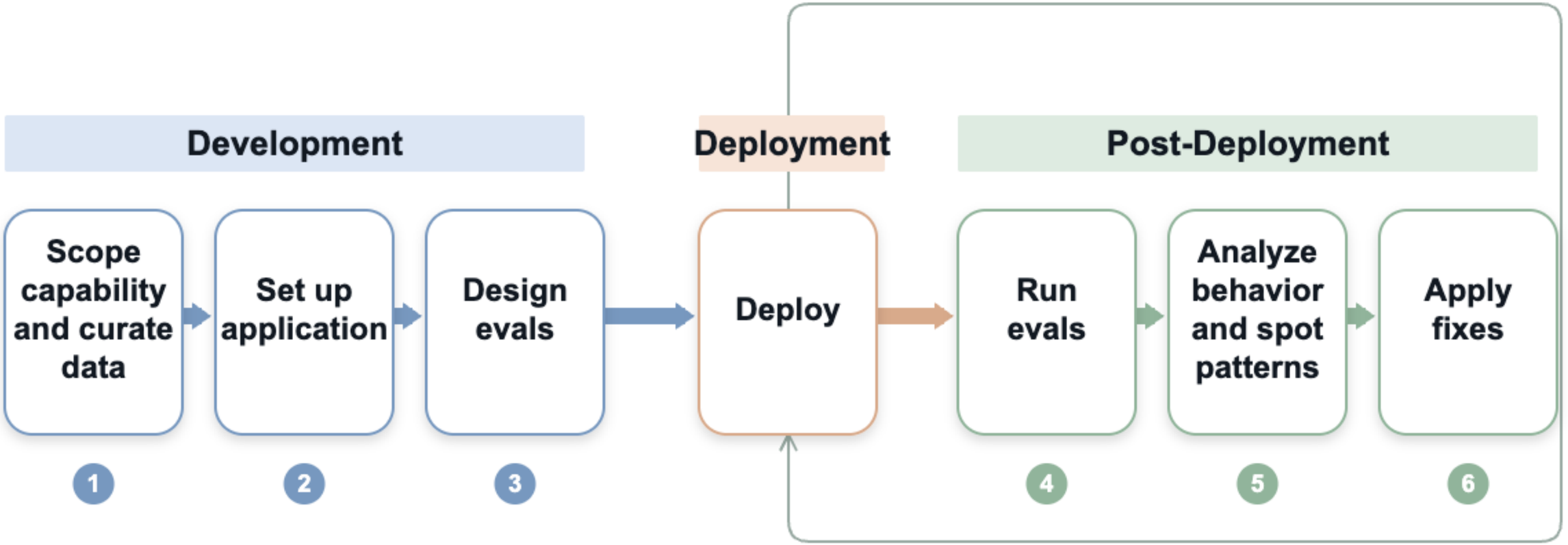
How do we deal with non-determinism?

- Use iterative system design: build, observe behavior, improve, rebuild.
- Example: a coding agent for engineers.
  - Wrong edits, missed context, inconsistent style, flaky test fixes.
  - Patch the workflow, update prompts and tools, improve evals.
- Keep iterating until the experience is reliably good.



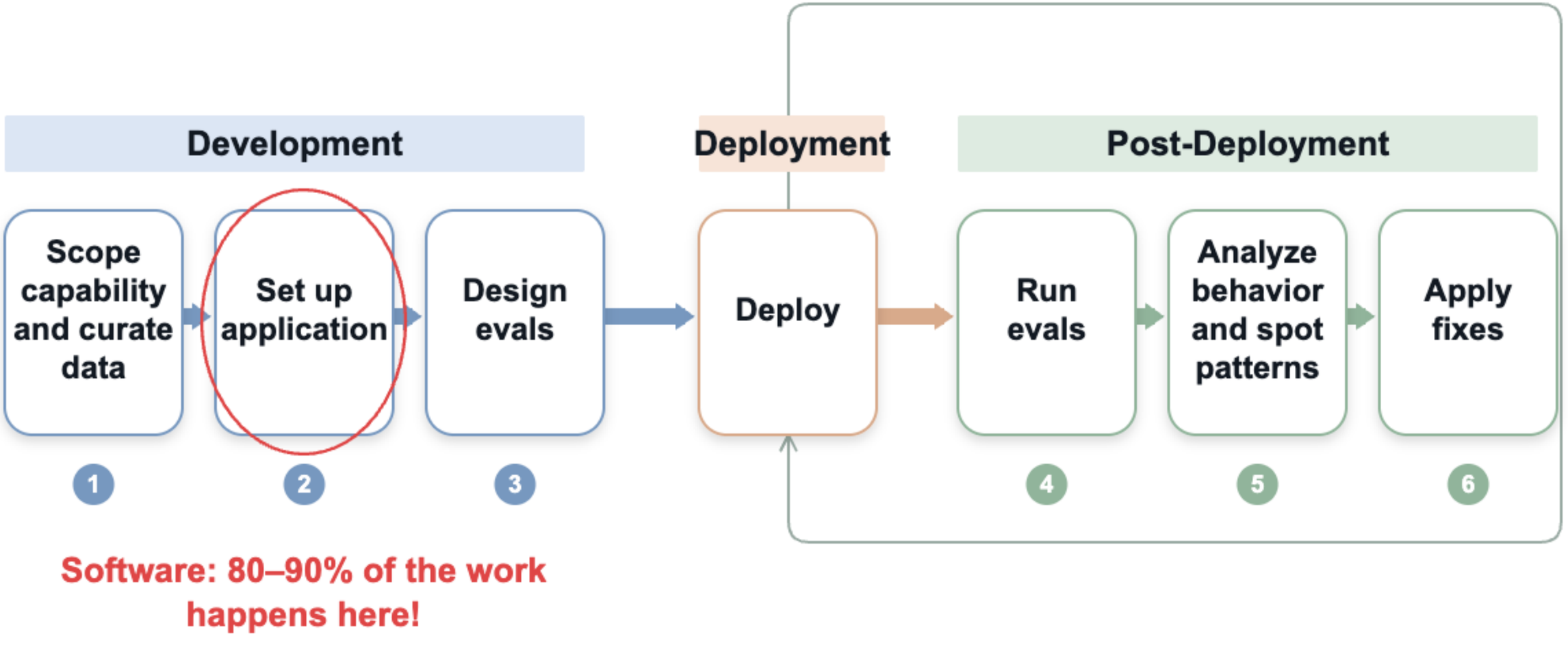
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This is an iterative process



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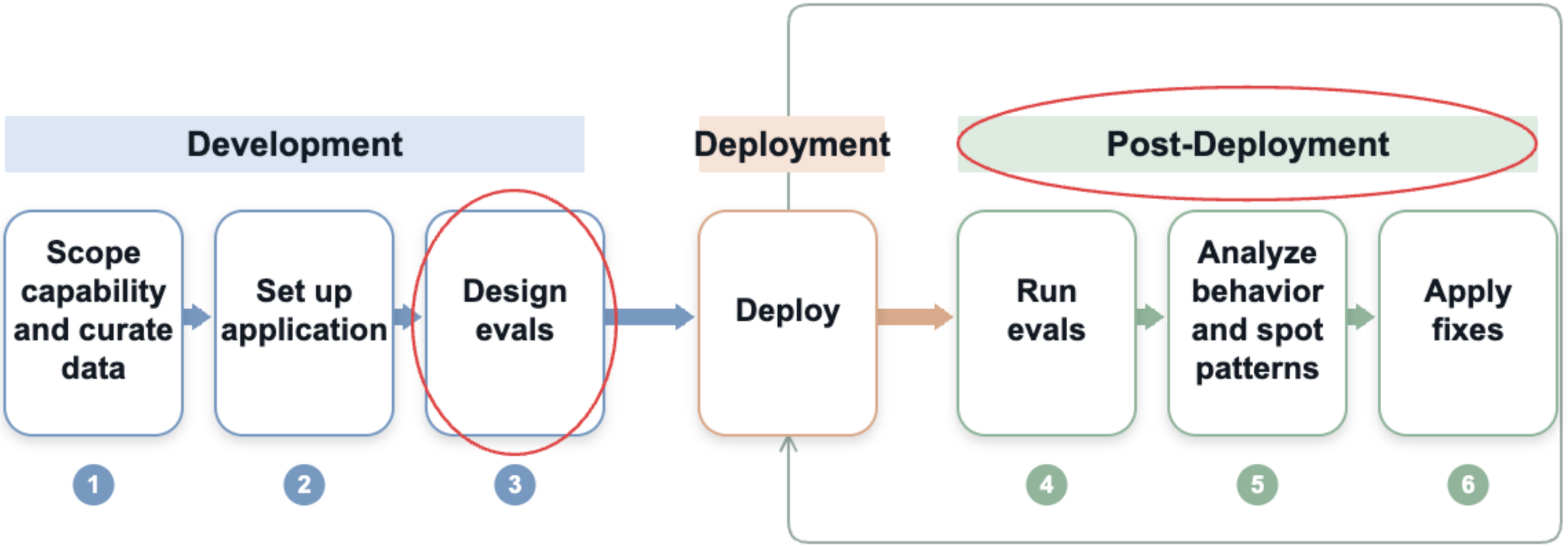
Traditional Software



Software: 80–90% of the work happens here!

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AI Applications



80–90% of the work happens here!

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LevelUp Lens™

We call this the 80/20 flip: in AI applications, most of the work shifts from building the first version to evaluating, understanding, and improving its behavior.

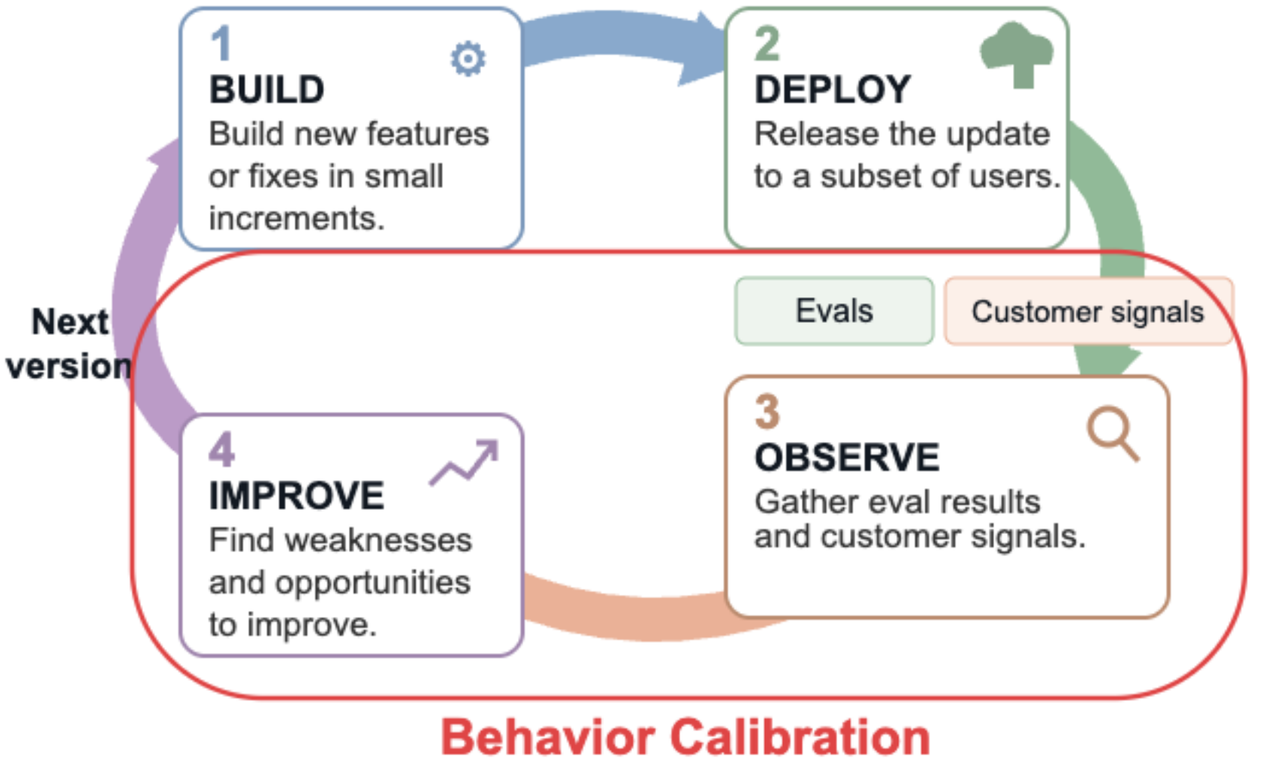
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Iterative System Design

For AI products, a lot of your time gets spent in **behavior calibration**:

- Did users **behave differently** than you expected?
- Did the AI respond in a way that's **calibrated to your product's expectations?**

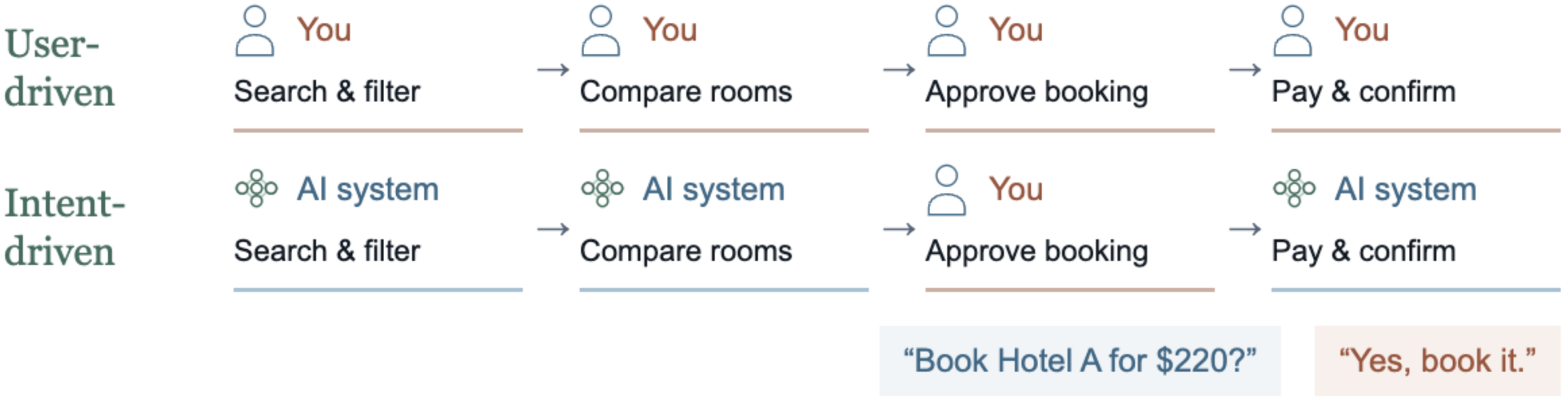
That's how you tame non-determinism.



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Problem 2: From user-driven to intent-driven workflows

> "Book a hotel in Paris next week, under $250, near the center."



Agency is the authority to take actions toward a goal.

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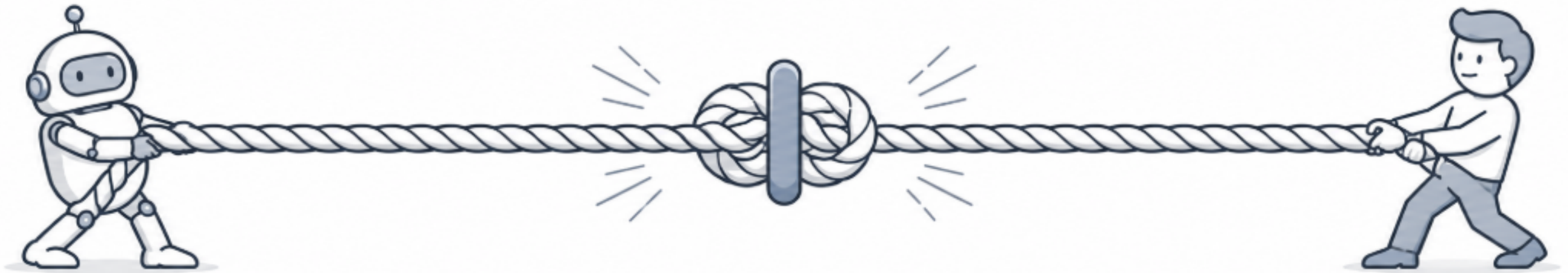
The agency-control trade-off

Agency

Decisions delegated to AI.

Control

Decisions retained by the person.



Choose the balance for each part of the system.

Adapted from our original Lecture 2 illustration

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An actionable way to deal with the agency-control trade-off

CC/CD

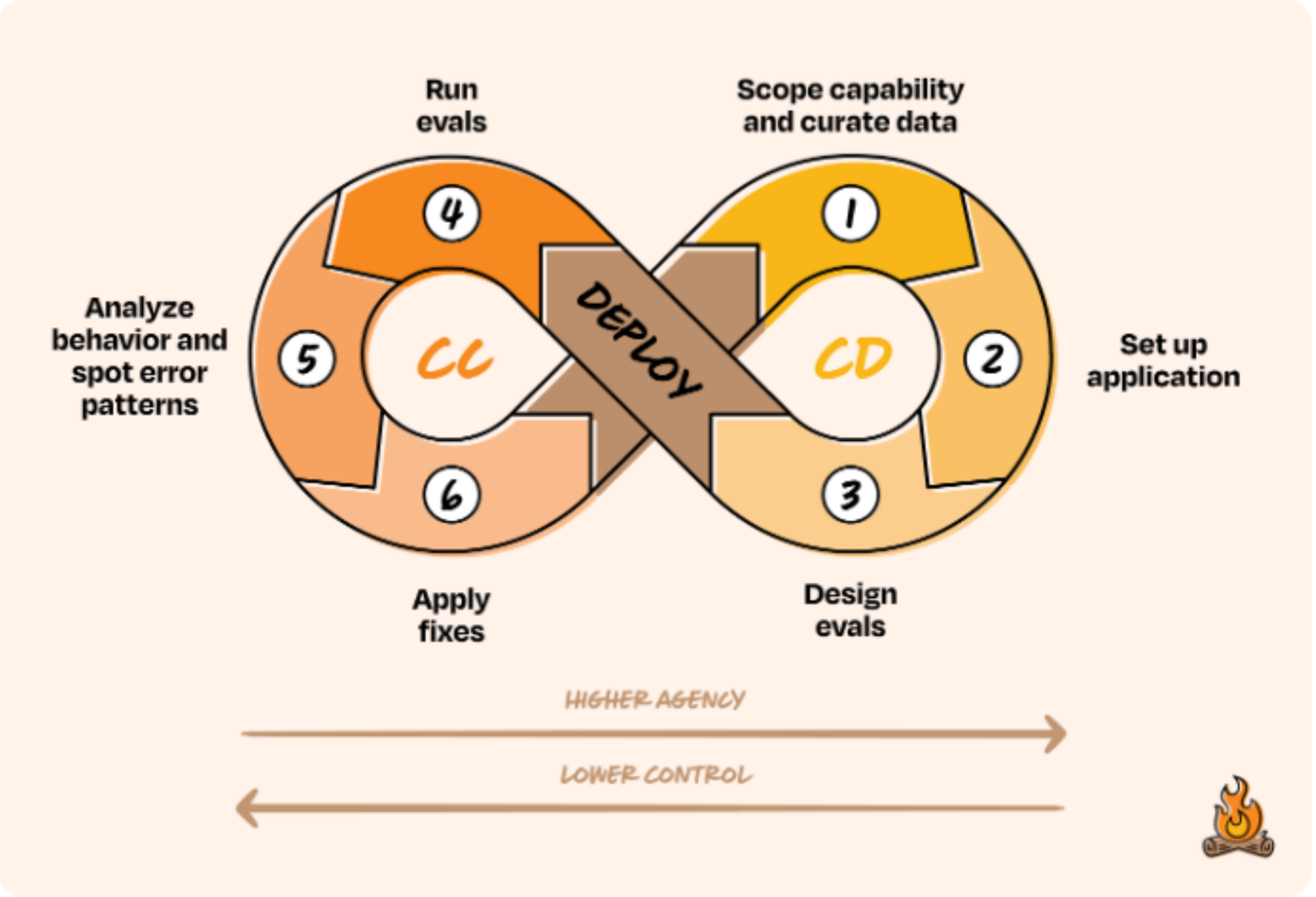
Continuous Calibration / Continuous Development

Improve reliability and calibrate which decisions AI can make.

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CC/CD

Continuous Calibration / Continuous Development



Source: Aishwarya Reganti & Kiriti Badam, Lenny's Newsletter

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CC/CD: Continuous Development

1. ## Scope capability and curate data

   Define what this version should do, how much agency it has, and the data that describes expected behavior.

2. ## Set up application

   Build the simplest version, log its behavior, and make it easy for people to take back control.

3. ## Design evals

   Define how you will judge success and test the application against your reference data.

Source: Aishwarya Reganti & Kiriti Badam, Lenny's Newsletter

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CC/CD: Continuous Calibration

4 **Run evals**

Evaluate real interactions to understand how the deployed application is performing.

5 **Analyze behavior and spot error patterns**

Review failures and find recurring gaps between expected and actual behavior.

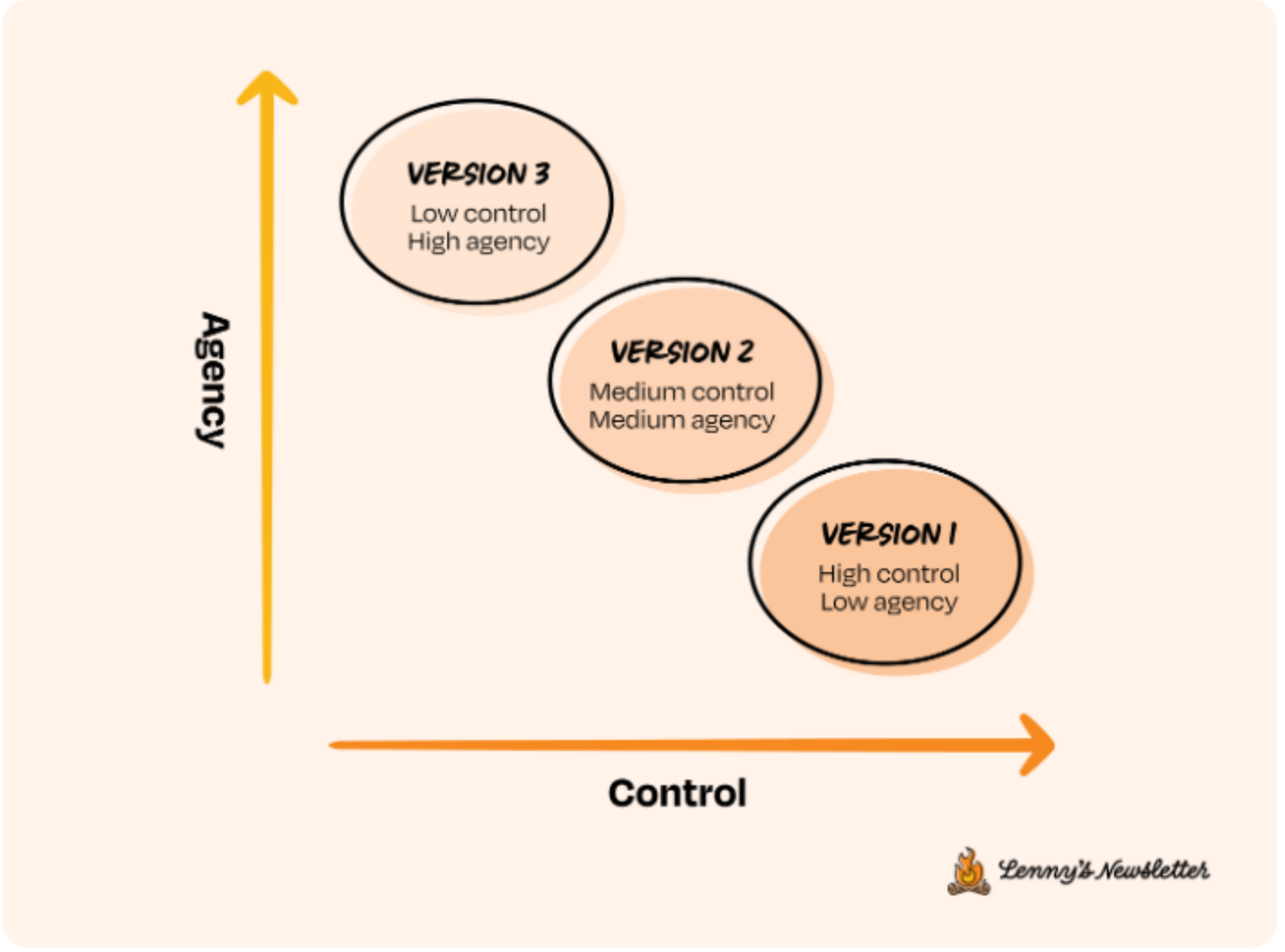
6 **Apply fixes**

Make targeted changes, re-run evals, and refine the application or the evals based on what you learn.

Source: Aishwarya Reganti & Kiriti Badam, Lenny's Newsletter

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CC/CD: learn from each version



V1: Route tickets

People resolve them.

V2: Draft resolutions

People review and act.

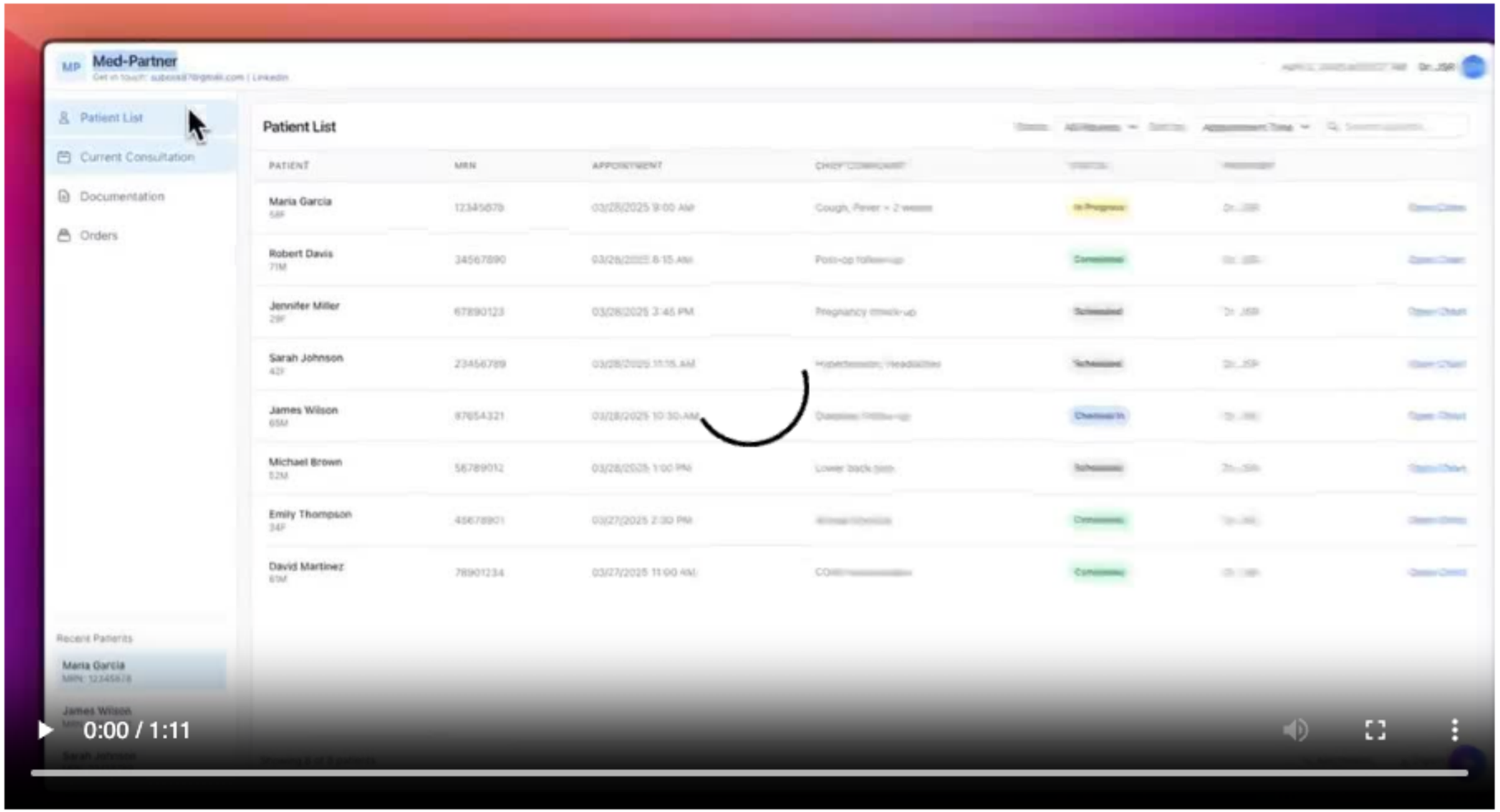
V3: Resolve scoped tickets

Hand exceptions back.

Source: Aishwarya Reganti & Kiriti Badam, Lenny's Newsletter

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Student demo: AI-powered medical assistant



What would you automate, assist, or keep under the doctor's control?

Source: Student project, original welcome deck, page 24

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Problem 3: Operational issues

- Hallucinations
- Sycophancy
- Bias reinforcement
- Latency
- Prompt sensitivity
- Inconsistent response length
- Limited explainability

We'll address these through evals and design patterns throughout the course.

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Let’s take a 3-minute break

What’s one thing you learned or unlearned so far?

Leave your comments in the chat.

Give a thumbs up 👍 to the answers you like.

The most-voted answer or the instructor’s favorite **bags 4 points.**



“I need a pee break.”

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From strategy to execution

How do we build the first version of a customer-support application?



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From generative AI to agentic AI

A quick recall from Lecture 0: what are we building?

Generative AI



+

Tools / APIs

Run code; call a service

External memory

Retrieve information for the task

Planning

Choose and revise the next steps

=

Agentic AI

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Building the first version

1

Context Design

What information and instructions does the application need?

→

2

Choose the right model

What does this task need the model to do?

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Context Design

V1: Route customer-support tickets

Read a ticket and choose the right support queue.

People still investigate and resolve the issue.



What information does this version need to route a ticket well?

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Define the data for your V1

Data = Input Context/Behavior + Expected Behavior

What the system sees

The ticket, relevant details
and the customer's situation.

What it should do

The correct queue and
the facts that justify it.

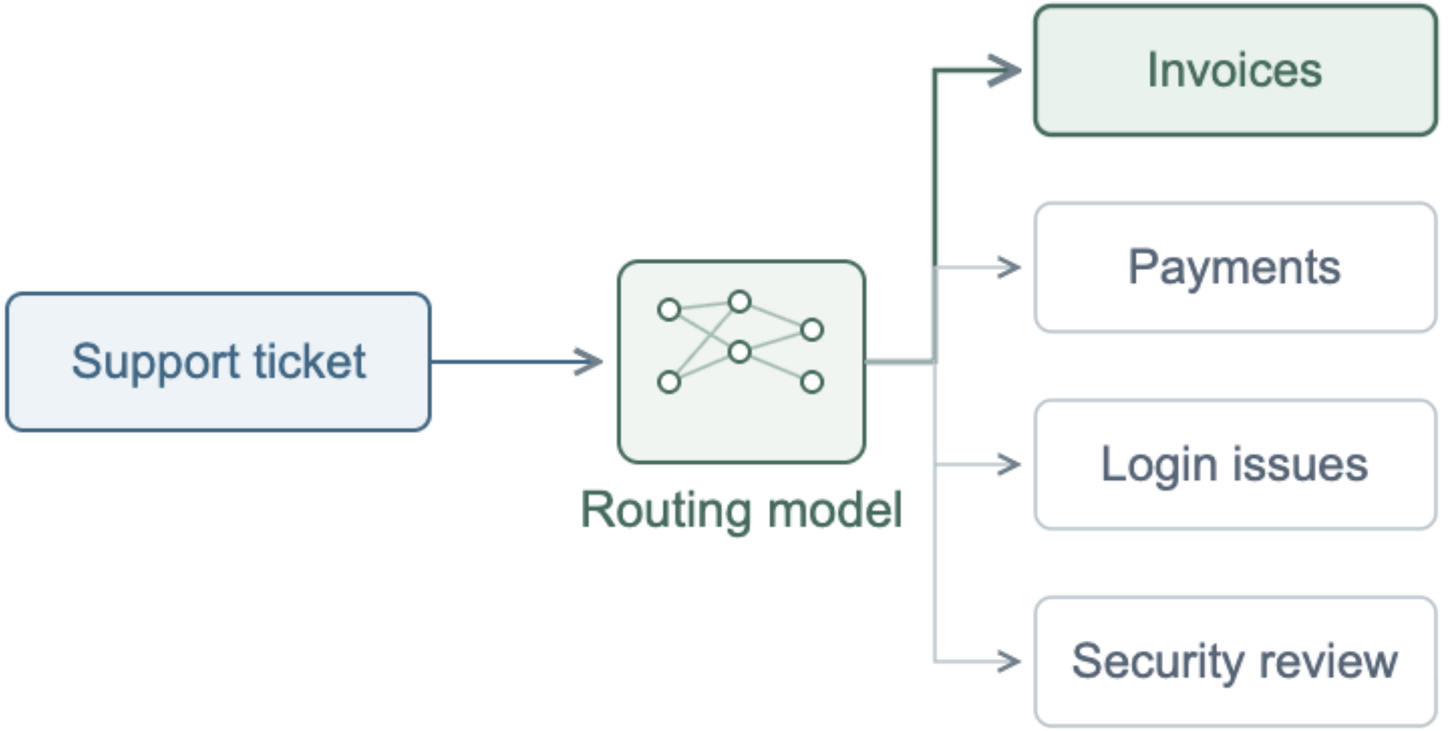
Start with **50–60 representative examples**.
Include clear cases, ambiguous cases and exceptions.

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When the ticket is enough

> Customer ticket
>
> “Can you send me last month’s invoice?”

Expected queue: **Invoices**



The request itself gives us enough information to choose the queue.

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When the ticket needs metadata

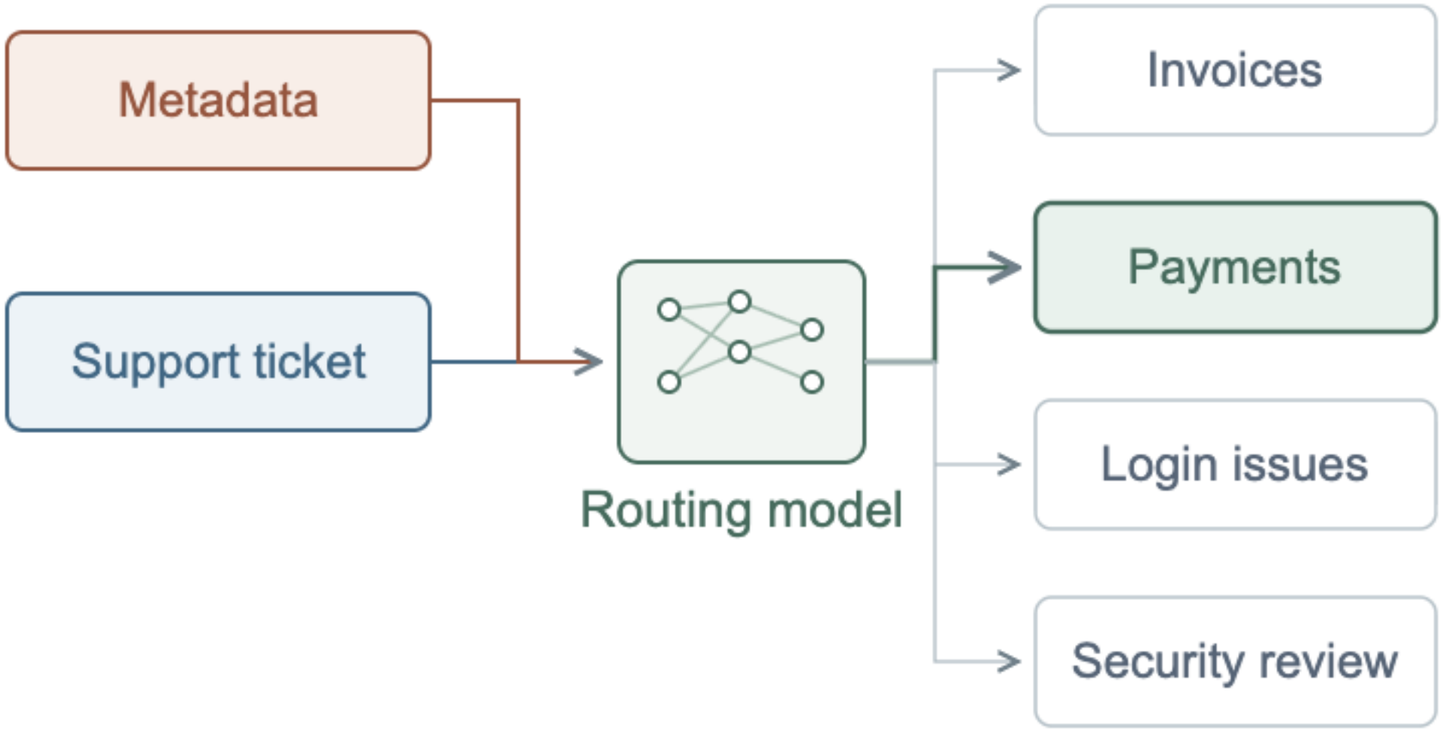
Customer ticket

"Payment deducted, but no order confirmation."

Transaction and order details

Payment: pending
Order: not found

Expected queue: **Payments**



The ticket raises the question; transaction data helps us route it.

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When the ticket needs customer history

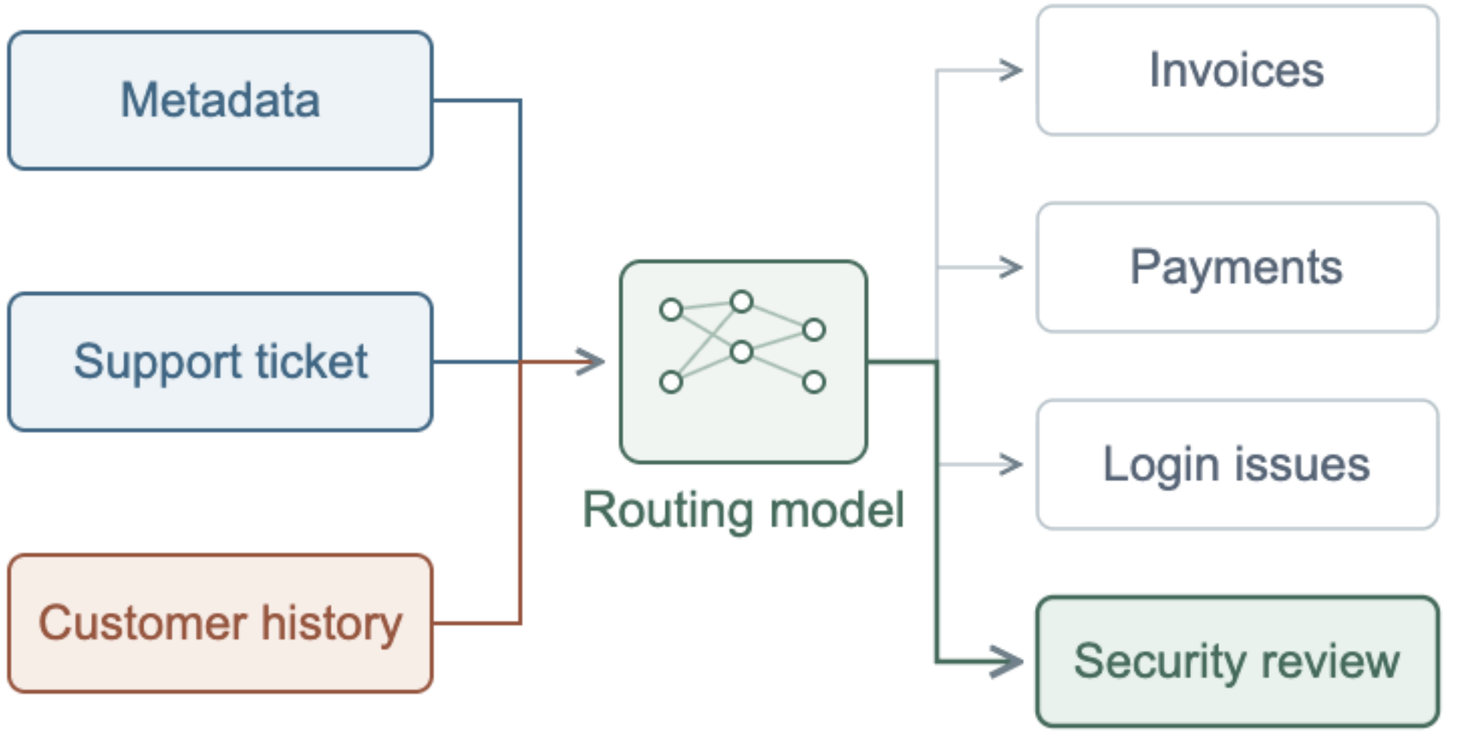
Customer ticket

"I keep getting locked out, even after resetting my password."

Previous tickets

2 recent lockouts. Password resets did not help.

Routing rule: recurring lockouts go to **Security review**.



The history changes how we interpret the same complaint.

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Context Design = Architecture for AI

What the examples revealed	Context needed	Application requirement
A clear invoice request	Ticket content	Read the support ticket
An unclear payment issue	Transaction + order status	Connect to those records
Repeated login problems	Past tickets + routing rules	Retrieve history and apply policy

Add each source because your examples show why the V1 needs it.

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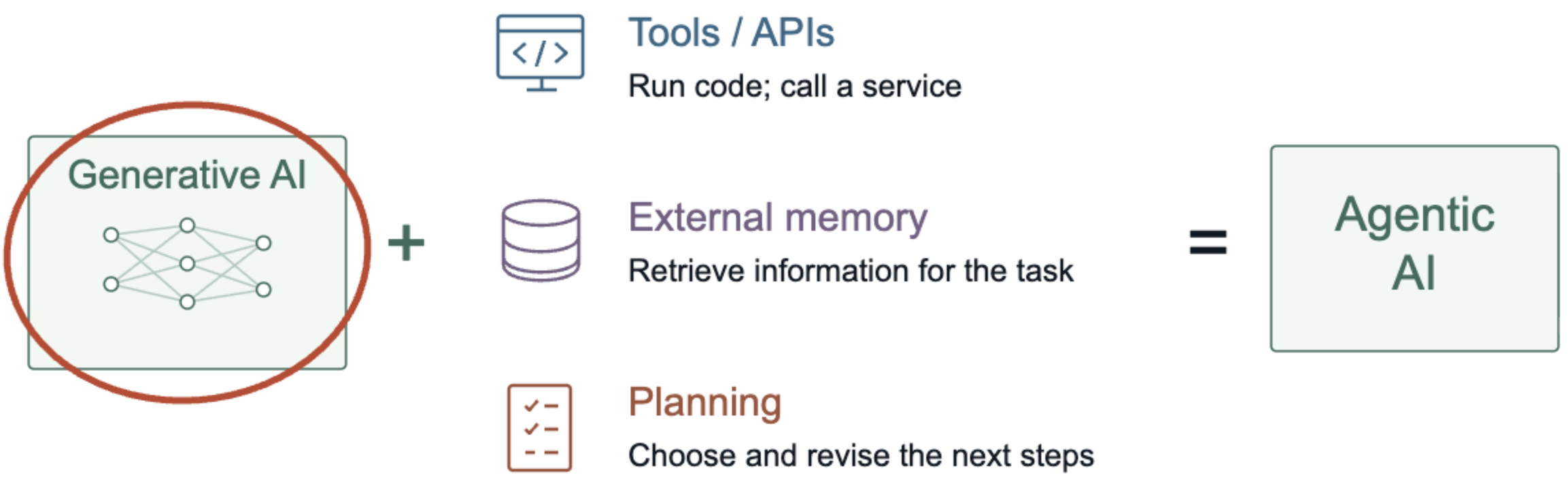
LevelUp Lens™

Your dataset should guide your application design.

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Choosing the Right Model

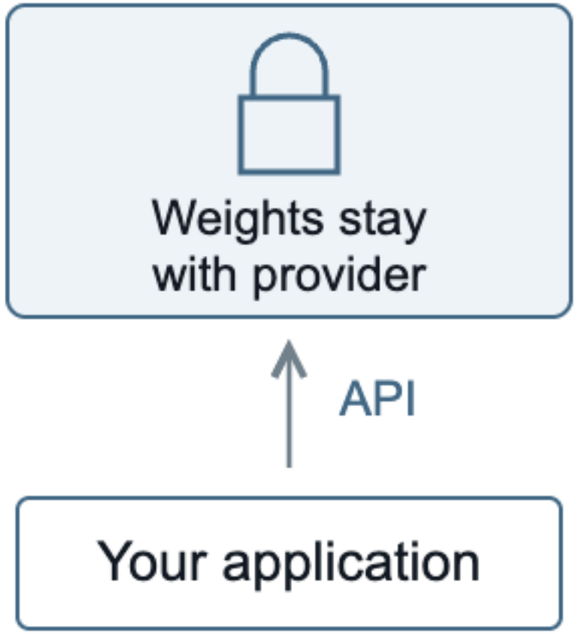
We have designed the context. Which model should use it?



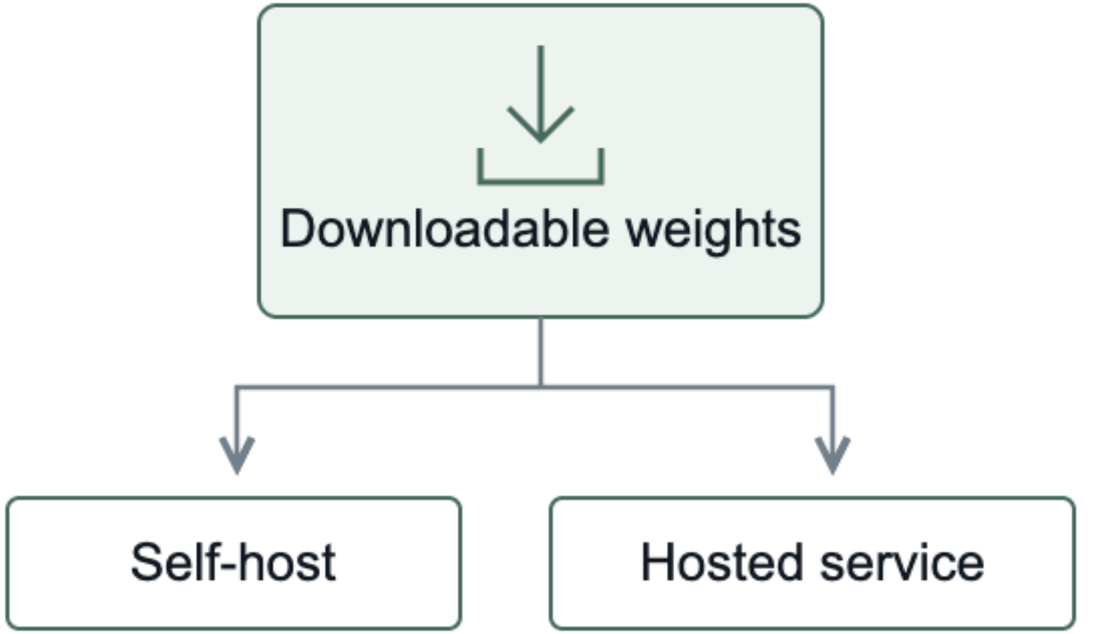
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Open and Closed Models

Closed model



Open-weight model



Open weights do not mean you have to host the model yourself.

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Where to Compare Models

Arena AI ↗

Compare model responses and human preferences.

Look at the categories relevant to your task.

Artificial Analysis ↗

Compare benchmark results, speed and pricing.

Check the model version and reasoning configuration.

Use these to shortlist candidates, then test on your own examples.

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Why Start with a Hosted Model?

Less infrastructure

The provider runs the model, manages GPUs and handles serving.

Faster experiments

Try different models without deploying each one yourself.

Focus on your V1

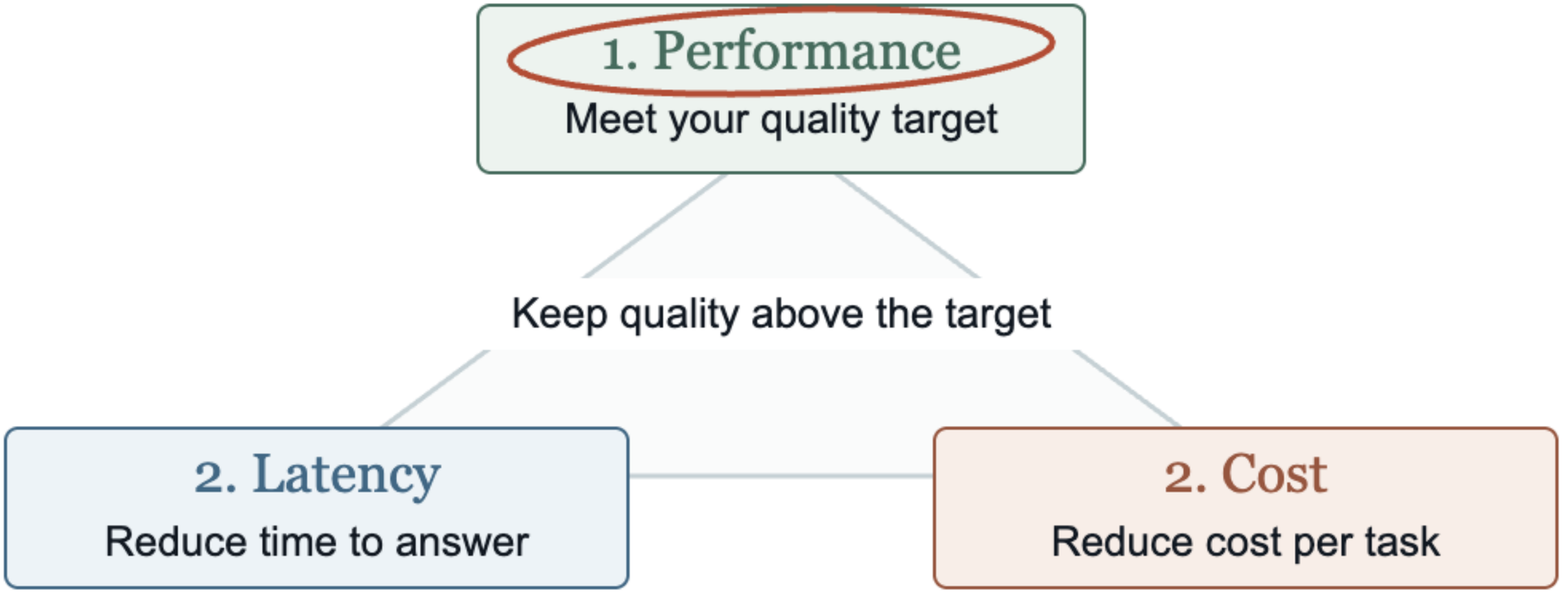
Spend time on context, instructions and testing the behavior.

A hosted model can be closed or open-weight.

Check data requirements and provider terms before choosing.

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Optimize Performance First

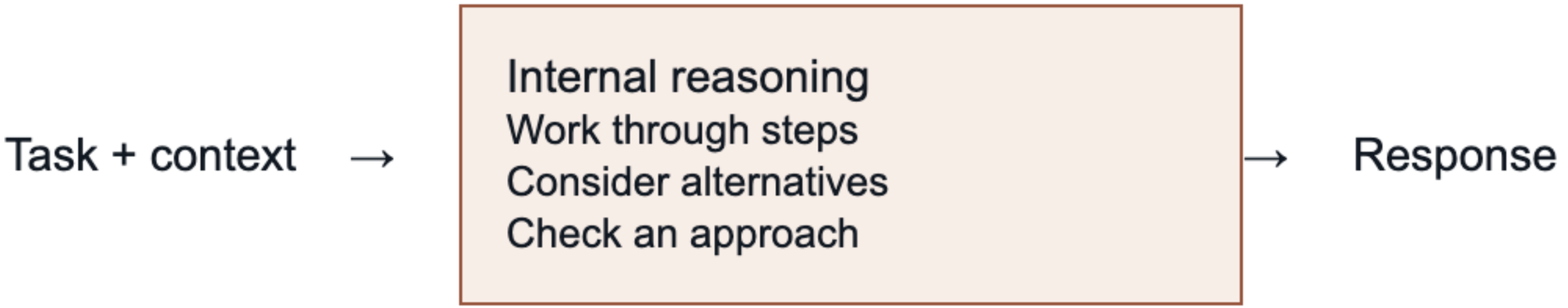


Then test a smaller model or less reasoning, and measure again.

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Note on reasoning models

Reasoning is now a standard capability in leading general-purpose models.



You may have seen this exposed as a “thinking” or “reasoning effort” setting.

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What Changes When You Increase Effort?

The same model can spend more computation working through the task.



More effort can improve difficult answers, but usually adds time and cost.

OpenAI: how reasoning works ↗ · Anthropic: effort ↗

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Don’t fall into model selection hell: start with a mid-tier model and low reasoning effort, then adjust based on failures in your own tests.

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Reasoning and Model Choice as Levers

Reasoning as a lever



Give the same person more time to work through it.

More computation while answering.
The learned model stays the same.

A different model as a lever



Give the problem to someone with different skills.

Different learned capabilities.
Size and training both matter.

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Increase Reasoning or Change the Model?

Keep the task, data and instructions fixed.

Increase reasoning

It understands the pieces but struggles to connect the steps.

Policy exceptions, multi-step math, or planning with constraints.

Try another model

It needs different language, visual or domain capabilities.

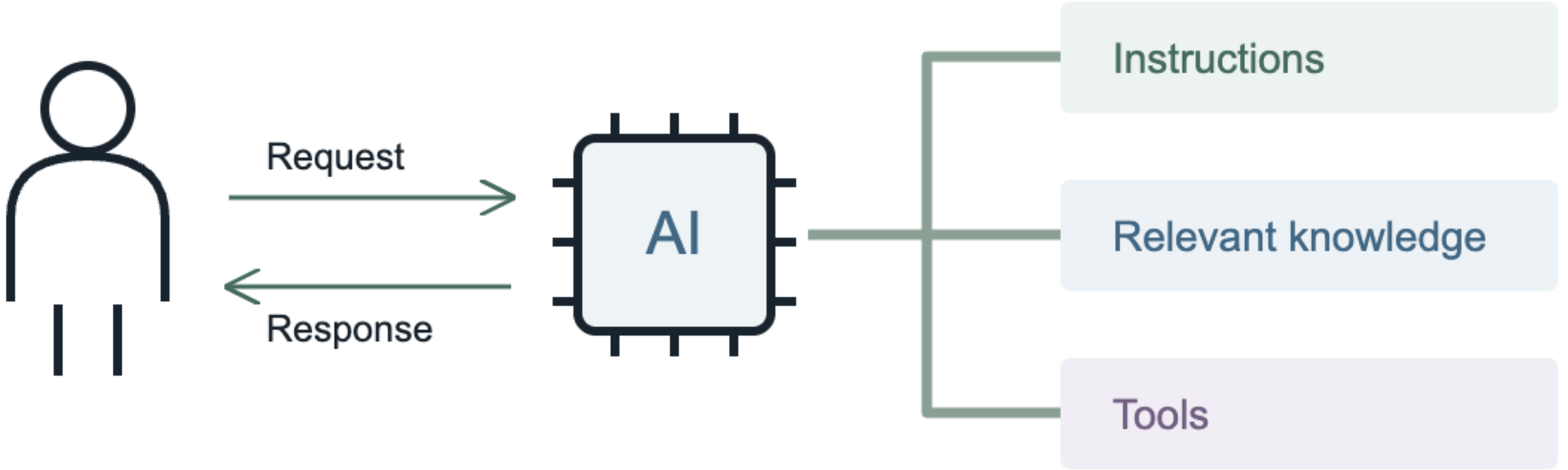
Robotic writing, misread drawings, or missed domain concepts.

Compare quality, latency and cost on the same examples.

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From Context Design to Context Engineering

You have the data and a model. How do they work together?



Context engineering is the iterative plumbing that connects the model to what it needs.

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Context Engineering

Context Engineering = Instructions + Tools + External/Past Knowledge

What it should do

Route the ticket.
Follow the routing rules.

What it can use

Approved tools to retrieve the information it needs.

What it should know

Relevant policy, order details and customer history.

We will engineer each of these more deeply throughout the course.

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Instructions and Prompting

Context Engineering = Instructions + Tools + External/Past Knowledge

Prompting is how we give instructions to AI systems.

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System prompt vs user prompt

System prompt

Builder-provided instructions

Route tickets to Invoices, Payments, Login issues or Security review.

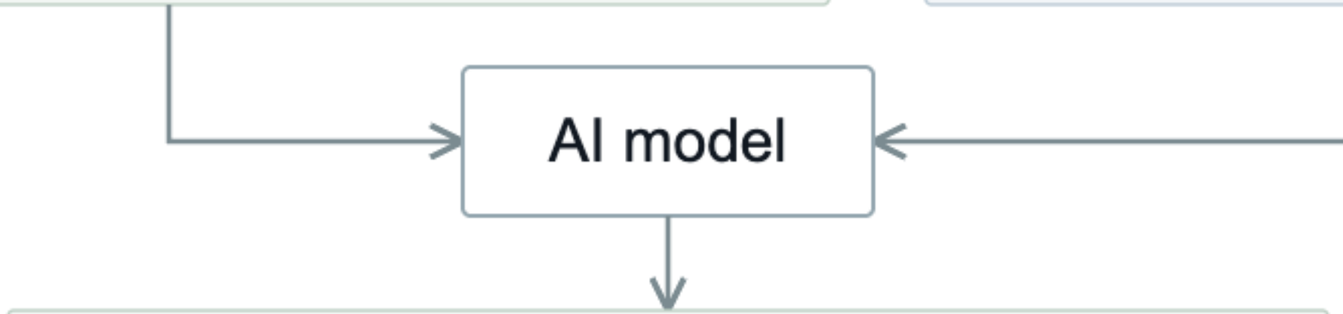
Return the queue and a brief reason.

Do not change accounts.

User prompt

Customer-provided request

Please send me a copy of my invoice.



Invoices · The customer requests an invoice.

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How prompting has evolved

Skill-based prompting

2022 onward

People write instructions and examples to guide the support router.

Level 1: In-context learning · Level 2: Advanced methods

Automated prompt optimization

2023 onward

A model or algorithm proposes prompt changes; tests show which work better for routing.

Level 3: Meta-prompting · Level 4: Algorithmic optimization

Self-optimizing models

Late 2024 onward

Models trained to reason through the task need less explicit reasoning guidance.

Level 0: Reasoning models

These developments overlap. All three approaches remain useful.

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Prompting a support router

> Customer ticket: Please send me a copy of my invoice.

Earlier chain-of-thought prompts

Route this ticket using the rules.
Think step by step:
identify the issue, compare
the categories, then decide.
Return the queue and
a brief justification.

Prompting a reasoning model

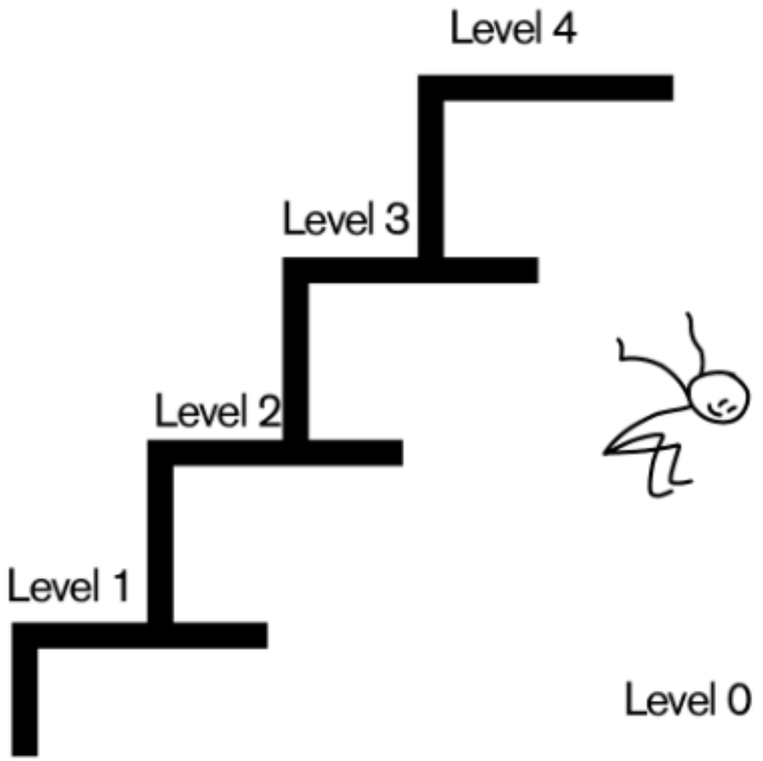
Route this ticket using the rules.
Return the queue and
a brief justification.

Illustrative output for both: Invoices · The customer requests an invoice.

Reasoning models often need less instruction about how to reason.

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Prompting today



Only three things to focus on

1. Define the task, context, and expected output.
2. Give examples if needed.
3. Follow your model's prompting guide.

   OpenAI ↗ Anthropic ↗

$$\text{End-user prompting skill required} \propto \frac{1}{\text{AI capabilities}}$$

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Meta Prompting Demo in Arize

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The End

What's one thing that you have learned or unlearned in this lecture?

Leave your comments in the chat.

Give a thumbs up 👍 to the answers you like.

The most-voted answer or the instructor's favorite **bags 4 points.**



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Logistics Details

Slides, lecture notes, and the recording will be available in 2–3 hours.

- Attend tomorrow's lecture or watch the recording.
- Complete the pre-course material if you haven't already.
- Make sure you're signed up on Slack, the education portal, and Maven.
- Add all course events to your calendar.
- Take the time to digest and come with questions to office hours.

Need help?

Post in our logistics channel on Slack:

#pfai_logistics_questions_october_2026

Don't have Slack access? Email:

problemfirst.ai@gmail.com

See you tomorrow!

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